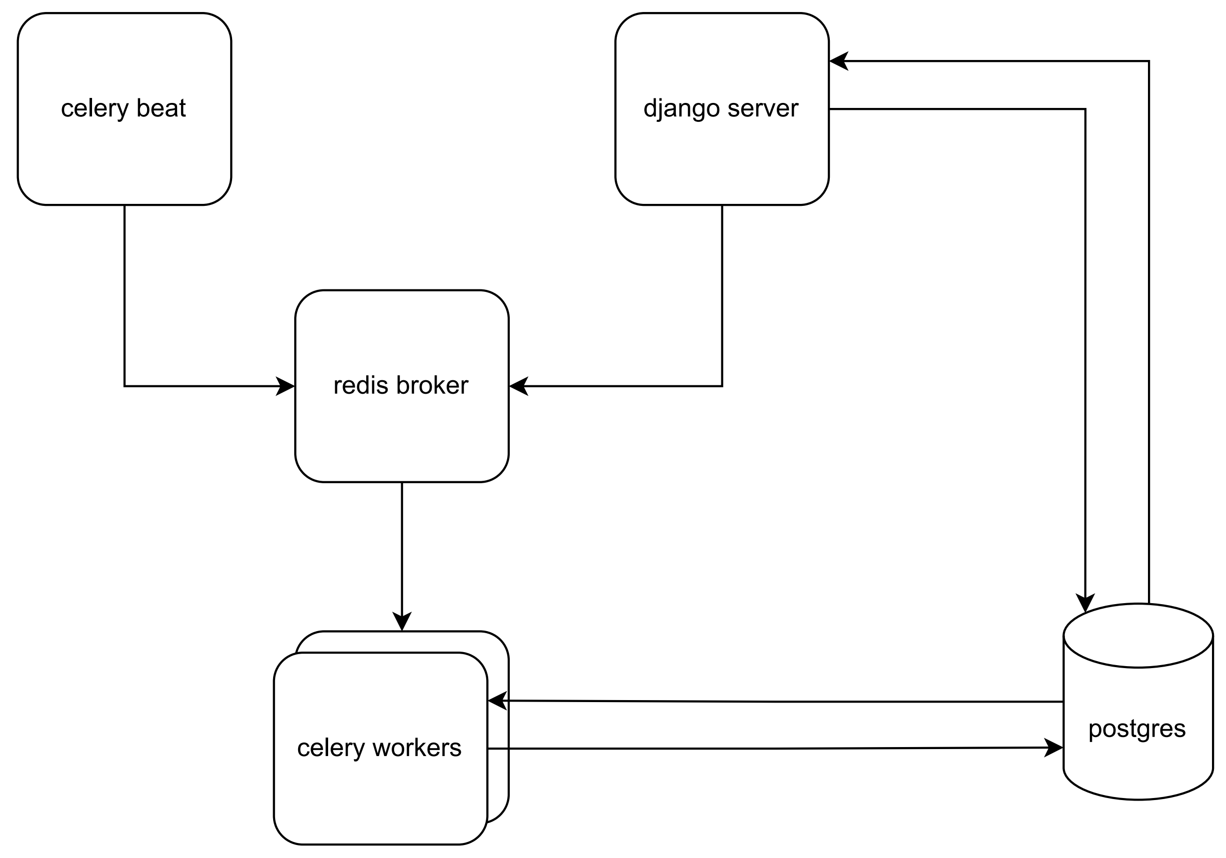
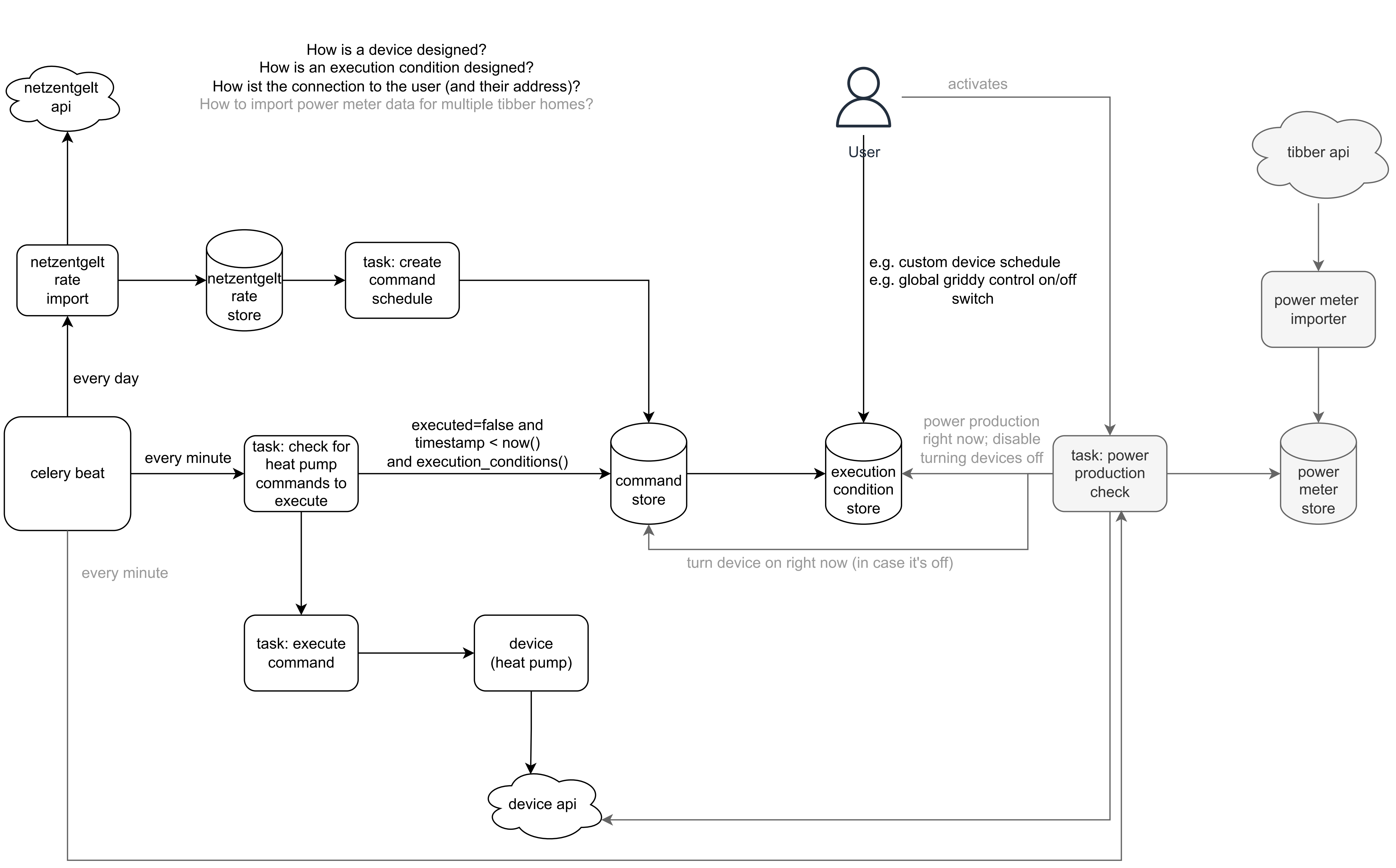
General Architecture



Griddy Control Process



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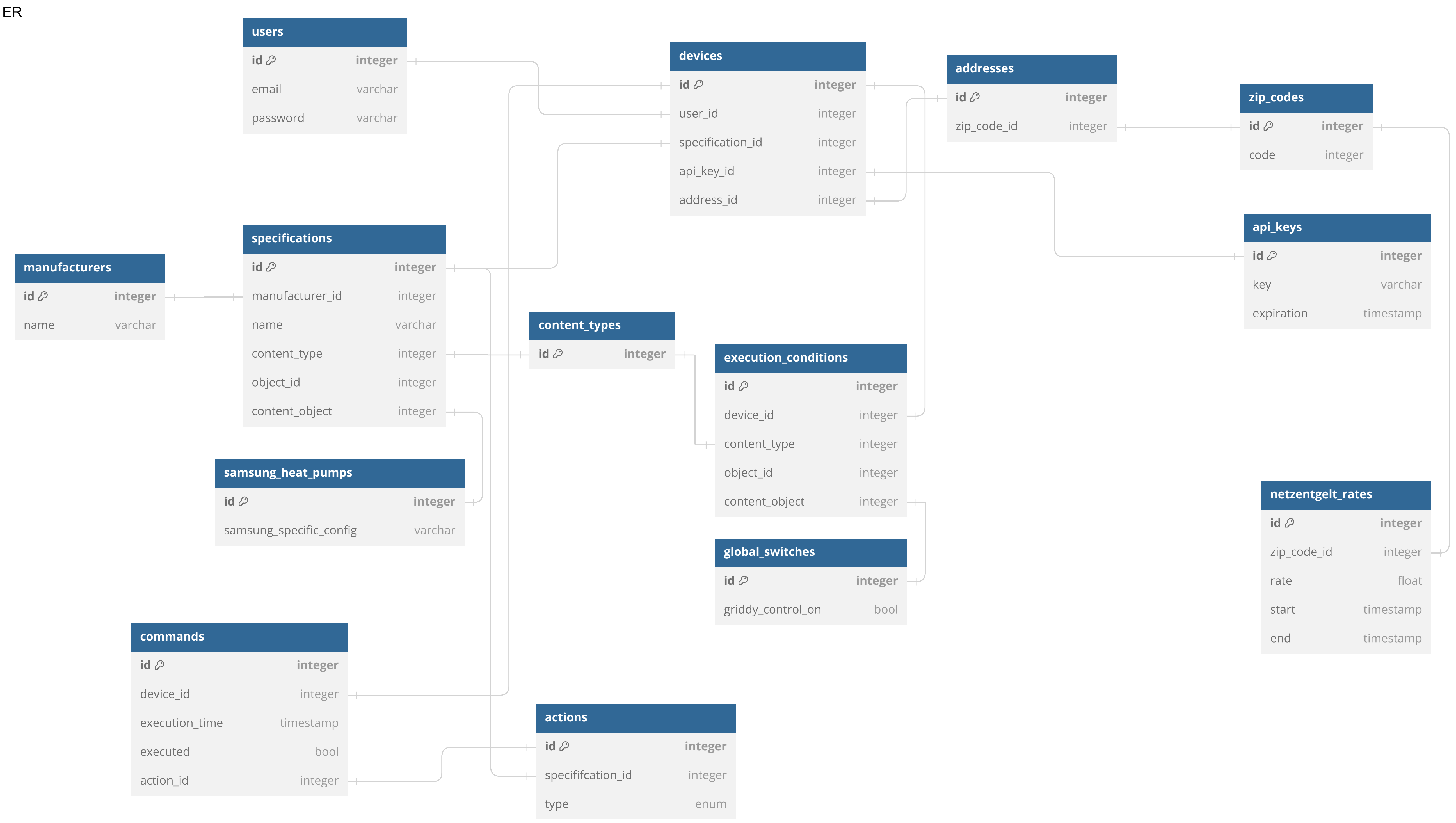

API for specific devices and execution conditions

```python
# E.g. SamsungHeatPump model must derive from this class
class DeviceImplementation:
    def execute_action(self, action: str):
        action_method = getattr(self, action, None)
        if not action_method:
            raise AttributeError(f"Action {action} not found on implementation {self.__class__}.")
        return action_method()


class HeatPumpModelXYZ(DeviceImplementation):
    def turn_on(self):
        print("Turning on heat pump.")

    def turn_off(self):
        print("Turning off heat pump.")

    def set_flow_temperature(self, temperature: int):
        print(f"Setting flow temperature to {temperature}°C.")
```

```python
# E.g. GlobalSwitch model must derive from this class
class ExecutionConditionImplementation:
    def check_condition(self):
        raise NotImplementedError("Method check_condition not implemented.")

class GlobalSwitchCondition(ExecutionConditionImplementation):
    def check_condition(self):
        print("Checking global switch condition.")
        return True
```

```python
def main():
    heat_pump = HeatPumpModelXYZ()
    heat_pump.execute_action("turn_on")
    heat_pump.execute_action("set_flow_temperature")
    heat_pump.execute_action("turn_off")

    condition = GlobalSwitchCondition()
    condition.check_condition()
```

django apps